

SAT Structural Hardness IIb

Local Frustration Fields and the Failure of Scalar Compression

Christof Krieg

MAAT Project – Independent Research

<https://maat-research.com>

2026

Abstract

Paper 50 showed that graph-local SAT diagnostics improve over density-only baselines, but also that scalar F_{MAAT} and global robustness scores are weak predictors of SAT hardness. The present companion note tests a sharper thesis:

SAT hardness is damaged by aggressive global scalar compression.

Instead of treating hardness as a function of a single global robustness projector, we construct local frustration fields, rare-event tail summaries, and hotspot-cluster diagnostics.

Using a fixed-seed synthetic ensemble of 550 formulas across random, planted, and modular 3-SAT families, we compare density-only, standard graph, global-MAAT, local-frustration, and combined raw graph/frustration models. The target remains a deterministic DPLL node-count proxy, so the result is not claimed to be solver-independent in the strong sense.

The test supports the thesis in a restricted but useful sense. Random-forest global-MAAT compression reaches only

$$R^2 = 0.1375,$$

whereas local frustration plus density and formula type reaches

$$R^2 = 0.2946.$$

The strongest combined raw graph plus frustration model reaches

$$R^2 = 0.3185.$$

Thus the useful hardness signal is not carried by scalar F_{MAAT} or R_{rob} , but by phase-aware graph-local frustration geometry. The result does not solve SAT hardness. It identifies where the previous MAAT compression loses information.

1 Purpose and Status

This paper is a companion test to Paper 50, not the next roadmap item after Paper 50. Paper 49 already assigned later work to quantum pointer-state selection and cross-domain transfer. The role of the present note is narrower:

explain why the SAT result in Paper 50 remains weak by testing whether SAT hardness is lost during global scalar compression.

The paper is not a proof of $P \neq NP$, not a SAT solver competition, and not a modern CDCL benchmark. The solver target is still a deterministic DPLL node-count proxy. The contribution is diagnostic: it tests whether local frustration fields retain more hardness information than F_{MAAT} and R_{rob} .

2 Thesis

Paper 50 suggested that the compression chain

$$d_a(x) \rightarrow \Gamma_a(x) \rightarrow R_{\text{rob}} \rightarrow F_{\text{MAAT}} \quad (1)$$

is too lossy for SAT. For field systems and smooth cosmological backgrounds, global robustness can act as a useful low-frequency stability projector. SAT hardness is different. It is controlled by local and rare-event structures:

- contradictory local clusters;
- forced-propagation traps;
- fragile backdoor structure;
- high-degree and polarity-imbalanced hotspots;
- short-cycle and modular frustration.

The working thesis is therefore

$$\text{Hardness} \not\sim f(\text{global robustness}), \quad \text{Hardness} \sim f(\text{multi-scale frustration geometry}). \quad (2)$$

3 Local Frustration Fields

For every variable v in the clause-variable graph we define a local frustration proxy

$$d_{\text{frust}}(v) = c_1 d_{\text{deg}}(v) + c_2 d_{\text{pol}}(v) + c_3 d_{\text{cycle}}(v) + c_4 d_{\text{prop}}(v) + c_5 d_{\text{contr}}(v) + c_6 d_{\text{backdoor}}(v). \quad (3)$$

The coefficients are phenomenological and fixed in the script. They are not fitted by the regression model.

From this local field we compute distributional and hotspot summaries:

- mean, standard deviation, maximum, and Gini coefficient;
- tail quantiles $q_{75}, q_{90}, q_{95}, q_{99}$;
- tail mass and tail mean;
- hotspot fraction;
- largest connected hotspot-cluster fraction;
- hotspot edge density.

We also define three SAT-specific robustness candidates:

$$R_{\text{tail}} = \frac{1}{1 + q_{95}(d_{\text{frust}})}, \quad (4)$$

$$R_{\text{var}} = \exp[-\sigma(d_{\text{frust}})], \quad (5)$$

$$R_{\text{cluster}} = \exp[-2C_{\text{max}}/n], \quad (6)$$

and

$$R_{\text{SAT}} = (R_{\text{tail}} R_{\text{var}} R_{\text{cluster}})^{1/3}. \quad (7)$$

These quantities are not introduced as new universal MAAT sectors. They are SAT-specific diagnostics designed to test whether rare-event structure matters more than global robustness.

4 Benchmark Design

The benchmark uses a fixed random seed, 51051, and generates 550 synthetic instances:

Formula family	Instances
Random 3-SAT	330
Planted 3-SAT	110
Modular 3-SAT	110

The variable range is $24 \leq n \leq 38$ and the clause-density range is

$$3.26 \leq \alpha = m/n \leq 5.18.$$

The satisfiable fraction is 0.7018. No run reaches the DPLL node cap.

The regression target is

$$y = \log(1 + \text{DPLL nodes}). \quad (8)$$

Models are evaluated by five-fold cross-validation using ridge regression and random forests.

5 Model Ladder

The comparison is designed as a compression ladder:

Model	Role
Density-only	phase-density control channel
Density plus formula type	controls random/planted/modular family effects
Standard graph	ordinary graph-feature baseline
Global MAAT scalar/supports	$F_{\text{MAAT}}, R_{\text{rob}}, H, B, S, V$ compression
Local frustration fields	distribution and hotspot summaries only
Local frustration plus density/type	phase-aware local frustration test
Raw graph plus frustration	strongest combined feature model

6 Results

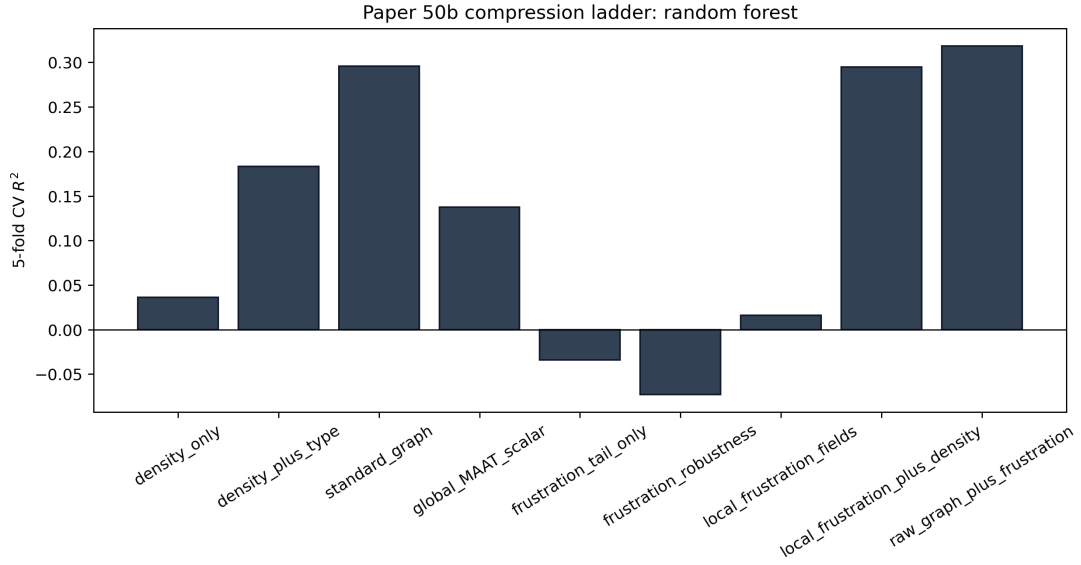


Figure 1: Random-forest compression ladder. Global MAAT compression is weak, while local frustration plus density/type nearly reaches the standard graph baseline.

Table 1: Random-forest cross-validated hardness prediction.

Model	R^2
Density-only	0.0362
Density plus formula type	0.1834
Standard graph	0.2959
Global MAAT scalar/supports	0.1375
Local frustration fields	0.0161
Local frustration plus density/type	0.2946
Raw graph plus frustration	0.3185

The key deltas are:

$$\Delta R^2_{\text{local+density-global MAAT}} = +0.1571, \quad (9)$$

$$\Delta R^2_{\text{local+density-density+type}} = +0.1112, \quad (10)$$

$$\Delta R^2_{\text{raw+frust-standard graph}} = +0.0226. \quad (11)$$

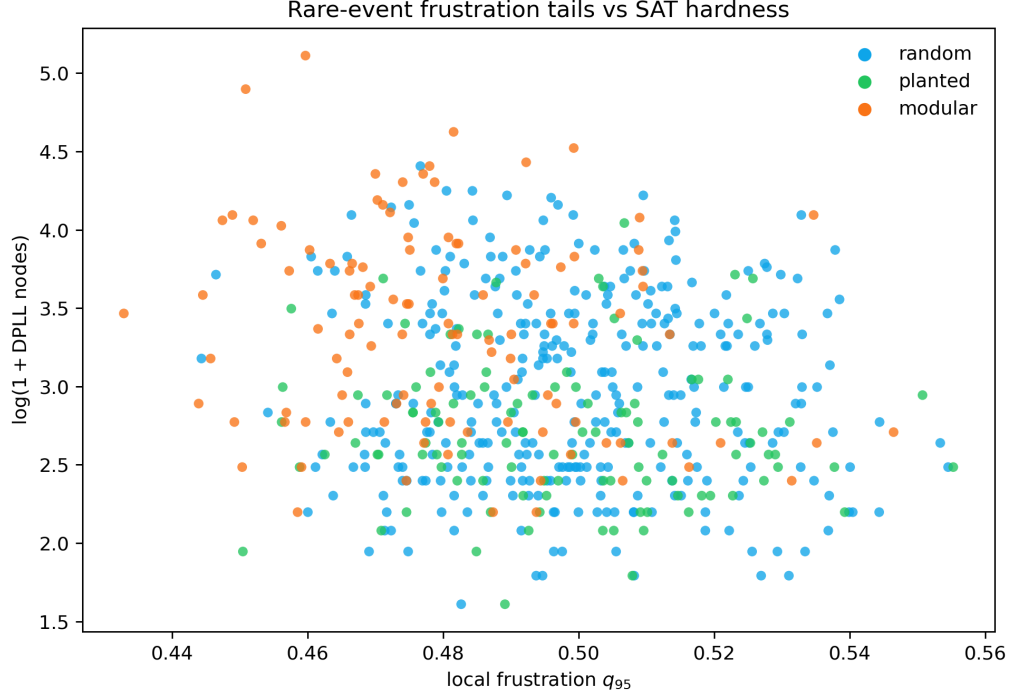


Figure 2: Local frustration tail q_{95} versus DPLL node count. The tail measure alone is not sufficient, but it contributes to the phase-aware local-frustration model.

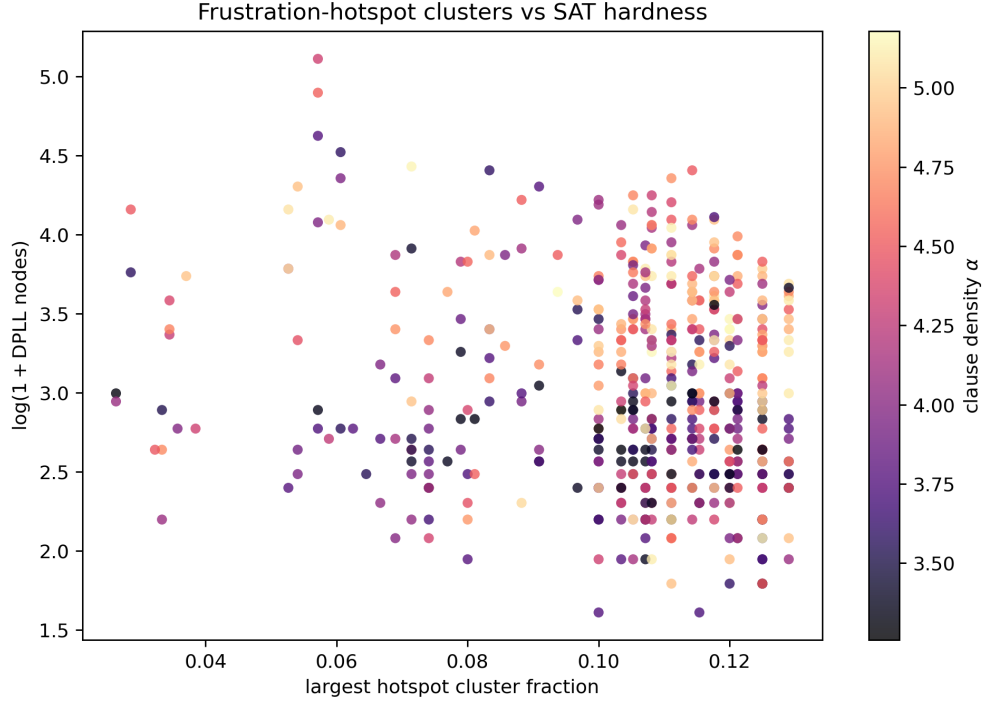


Figure 3: Largest connected hotspot-cluster fraction versus DPLL node count. Hotspot geometry is one component of the multi-scale frustration signal.

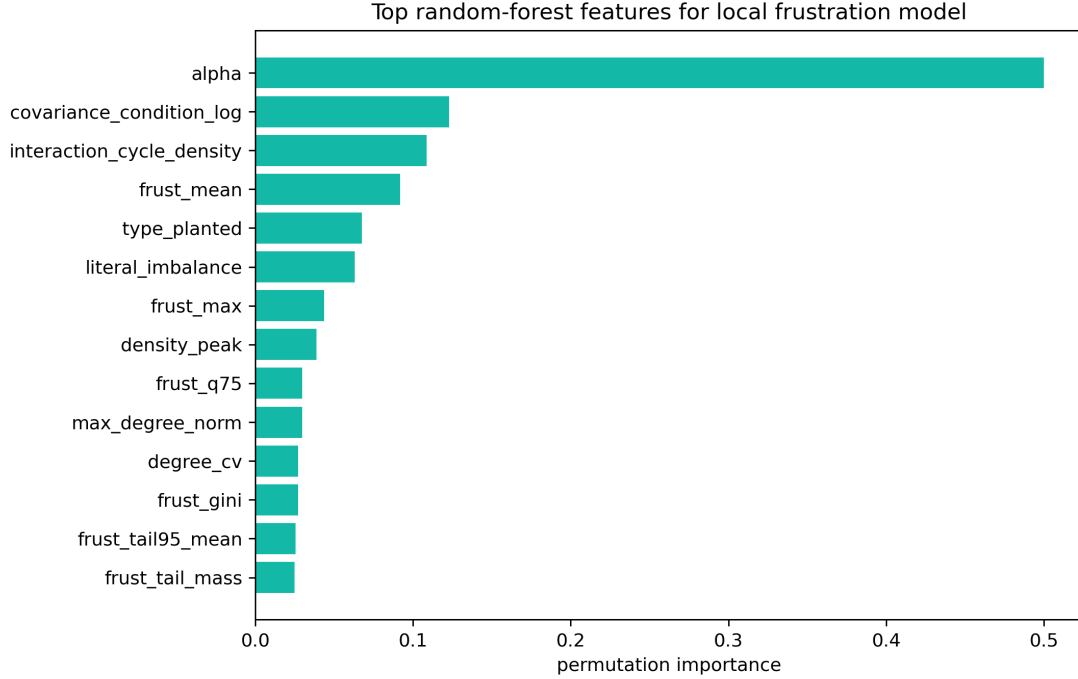


Figure 4: Top random-forest permutation importances for the combined raw graph and frustration model.

The top permutation-importance features are:

Feature	Permutation importance
α	0.4999
Covariance conditioning	0.1228
Interaction-cycle density	0.1088
Frustration mean	0.0918
Planted-family indicator	0.0676
Literal imbalance	0.0631
Frustration maximum	0.0438
Density peak	0.0388

7 Interpretation

The result supports the Paper-50 diagnosis. Scalar F_{MAAT} and R_{rob} do not capture SAT hardness well. The global compression acts like a low-frequency projection: it preserves broad coherence information but suppresses rare local structures.

However, the result is not a simple victory for the frustration variables alone. Local frustration fields by themselves remain weak. They become useful when combined with phase-density and formula-family information. The strongest model combines ordinary graph features and frustration fields.

Thus the correct conclusion is:

SAT hardness is graph-local, phase-aware, and tail-sensitive. The MAAT lesson is not that F_{MAAT} explains SAT, but that SAT requires a defect-field description rather than aggressive scalar compression.

8 Limitations

Several limitations remain:

- the target is deterministic DPLL node count, not CDCL conflicts, decisions, learned clauses, or runtime;
- the formulas are synthetic, not SAT Competition instances;
- the local frustration field is phenomenological;
- no graph neural network baseline is included;
- no external validation set is used;
- the strongest signal still depends heavily on density and formula family.

9 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/sat_frustration_fields_paper50b

Run:

```
python3 sat_frustration_fields_paper50b.py
```

The script writes:

- outputs_paper50b/paper50b_sat_instances.csv;
- outputs_paper50b/paper50b_model_comparison.csv;
- outputs_paper50b/paper50b_scalar_correlations.csv;
- outputs_paper50b/paper50b_feature_importance.csv;
- outputs_paper50b/paper50b_summary.json;
- outputs_paper50b/fig1_compression_ladder_rf.png;
- outputs_paper50b/fig2_tail_q95_vs_nodes.png;
- outputs_paper50b/fig3_hotspot_cluster_vs_nodes.png;
- outputs_paper50b/fig4_feature_importance.png.

The benchmark generates synthetic SAT formulas with a fixed random seed. No external SAT dataset is redistributed.

10 Conclusion

Paper 50b turns the SAT weakness from Paper 50 into a concrete test. The test supports a refined interpretation: SAT hardness is not governed by global robustness, but by graph-local, multi-scale frustration geometry.

This is not a final SAT theory. It is a correction to the MAAT compression strategy. For SAT-like domains, the relevant object is not a single scalar F_{MAAT} , but a structured defect field with tails, hotspots, and local frustration geometry.

References

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